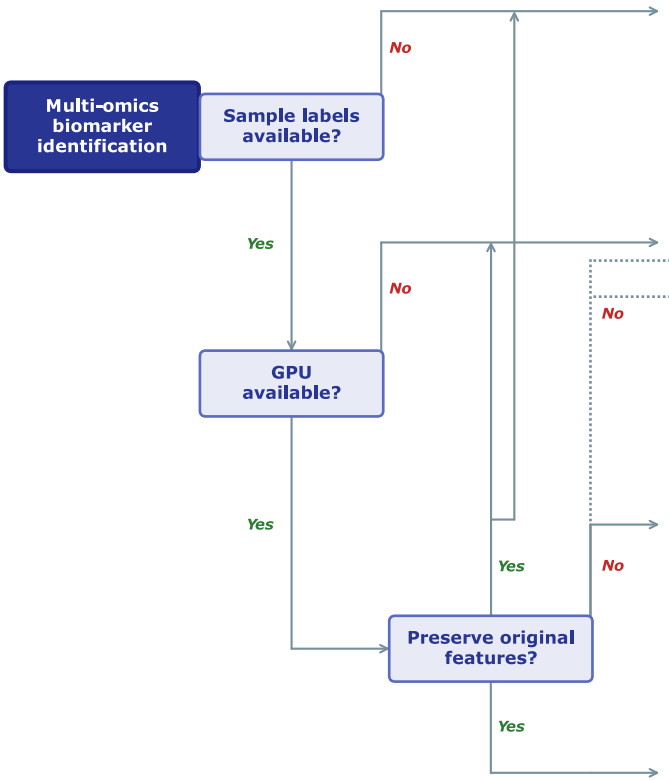

Method	Total time 100×40K	Total time 350×68K	Total time 650×85K	Peak memory 100×40K	Peak memory 350×68K	Peak memory 650×85K	Notes
GAUDI	53min	1h20min	1h48min	0.49 GB	2.6 GB	6.2 GB	Strong performance; unsupervised learning
MOFA	7min	11min	14min	0.39 GB	2.1 GB	4.9 GB	Variance-based loadings; weak performance
MCIA	<1min	8min	23min	0.49 GB	2.6 GB	6.2 GB	Variance-based loadings; weak performance
DIABLO	<1min	<1min	2min	0.39 GB	2.1 GB	4.9 GB	Strong performance; light computation
GAUDI	53min	1h20min	1h48min	0.49 GB	2.6 GB	6.2 GB	Strong performance; unsupervised learning
asmbPLS-DA	<1min	<1min	<1min	0.29 GB	1.6 GB	3.7 GB	Light computation; moderate performance
Stabl	15min	53min	1h48min	0.98 GB	1.2 GB	1.9 GB	Bootstrap sampling-based; moderate performance
GDF	16min	24min	32min	0.39 GB	2.1 GB	4.9 GB	Moderate performance; biased toward mRNA/miRNA/DNAm
DPM	<1min	2min	2min	0.29 GB	1.6 GB	3.7 GB	Univariate testing; weak performance
MOFA	7min	11min	14min	0.39 GB	2.1 GB	4.9 GB	Variance-based loadings; weak performance
MCIA	<1min	8min	23min	0.49 GB	2.6 GB	6.2 GB	Variance-based loadings; weak performance
CPU-only							
GPU required							
DeePathNet	3min	11min	13min	3.9 GB	11.7 GB	21.9 GB	Strong performance; broad omics coverage
DeepKEGG	<1min	<1min	<1min	0.07 GB	0.11 GB	0.12 GB	Strong performance; broad omics coverage; light computation
Pathformer	2min	10min	20min	79.3 GB	87.1 GB	102.5 GB	High stability; moderate accuracy, may exceed on weak signals
GENIUS	7min	1h8min	1h14min	1.4 GB	2.5 GB	4.3 GB	CNN with limited long-range context; moderate performance
P-Net	7min	39min	46min	0.27 GB	0.37 GB	0.41 GB	Low performance; sparse connectivity limits expressiveness
GNN-SubNet	2h25min	2h40min	2h56min	0.65 GB	0.83 GB	0.92 GB	Low performance; heavy computation; local message-passing
MOGLAM	2min	6min	7min	16.8 GB	65.2 GB	70.9 GB	Strong performance; highest stability
CustOmics	2min	4min	5min	2.6 GB	3.2 GB	3.6 GB	Moderate accuracy; unstable; autoencoder-based
TMO-Net	4min	44min	48min	9.1 GB	14.5 GB	18.8 GB	Moderate accuracy; unstable; autoencoder-based
MOGONET	3h42min	9h17min	10h13min	0.52 GB	0.75 GB	1.0 GB	Univariate feature ablation; slow; weak performance
MoAGL-SA	9h16min	16h	17h57min	0.55 GB	0.79 GB	1.6 GB	Univariate feature ablation; slow; weak performance
MORE	2h17min	4h7min	4h32min	0.21 GB	0.29 GB	0.52 GB	Univariate feature ablation; slow; weak performance